Physics-Guided, Identifiability-Aware Calibration of the Stanford RRAM Model for Sputter-Deposited TaO_x Devices

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***Abstract*—Compact-model calibration for resistive random-access memory (RRAM) is usually judged by the overlay between simulated and measured I–V curves, although a good overlay does not establish that fitted parameters are unique or physically supported. We present a reproducible calibration flow for the Stanford–PKU RRAM model that combines measured-cycle selection, conduction-regime classification, regime-conditioned priors, mechanism-aware loss weighting, HSPICE-in-the-loop Bayesian optimization, Fisher identifiability analysis, and cycle-level bootstrap uncertainty. The flow is applied to twelve sputter-deposited TaO_x conditions spanning 20–35% oxygen and 75–250 W sputter power. The full regime-aware method reproduces the measured DC butterfly curves with mean log-current RMSE of 0.530 decades for SET and 0.761 decades for RESET. Across 200 bootstrap resamples per condition, all active-parameter intervals are finite and nondegenerate, but the median coefficient of variation is 0.448; series resistance and velocity prefactors are especially unstable. Fisher analysis identifies only 0–8 of 16 active parameters per condition (mean 2.4), with condition numbers of 6.2×10^{18}–1.6×10^{66}. Series resistance and branch voltage/field scales are the most recurrently constrained coordinates, whereas current/gap decomposition and gap geometry remain poorly determined. Mechanism classification shows Schottky-dominant post-RESET high-resistance-state conduction in 92.5–100% of cycles for all twelve conditions, exposing a structural mismatch with the Stanford bulk current law. The resulting deliverable is therefore a behavioral compact- model calibration accompanied by explicit fit, mechanism, identifiability, and variability evidence.**

***Index Terms*—RRAM, TaO_x, compact model, Stanford model, identifiability, Bayesian optimization, conduction mechanisms, variability.**

I. Introduction

RESISTIVE random-access memory (RRAM) based on metal-oxide filamentary switching is a leading candidate for embedded nonvolatile memory and in-memory computing [1], [2]. Circuit- and architecture-level exploration of RRAM systems relies on compact models, of which the Stanford–PKU model [3], [4] is among the most widely used: it captures bipolar filamentary switching through a single gap state variable, a $\sinh$-type tunneling current, and a temperature-accelerated gap growth/rupture law.

A practical difficulty hides inside every reported "calibrated" parameter set. The Stanford model exposes more than twenty parameters, while a DC butterfly sweep constrains a handful of effective degrees of freedom: two resistance branches, two switching thresholds, and the curvature of each branch. Such over-parameterized models are *sloppy* in the sense of Transtrum *et al.* [5]: the data constrain a few stiff parameter combinations and leave the remaining directions to the optimizer's discretion. Two laboratories fitting the same measurement can therefore publish different "extracted" activation energies or gap geometries, each with an excellent overlay plot. Without an identifiability analysis, the reader cannot tell which numbers are measurements and which are artifacts of the prior or the search path.

This paper treats identifiability as a first-class deliverable of compact model extraction rather than an afterthought. We present an automated, physics-guided pipeline that fits the Stanford model to measured TaO_x DC switching data and accompanies every fitted vector with (i) a Fisher/Cramér–Rao report and (ii) cluster-bootstrap confidence intervals over measured cycles. The pipeline is applied uniformly to twelve TaO_x deposition conditions spanning a face-centered factorial design in oxygen flow and sputter power, so that the same evidential standard applies to every extracted parameter set.

The specific contributions are:

1) *Physics as a front-end diagnostic.* Every resistance-state segment of the representative switching cycle is classified into ohmic, Poole–Frenkel (PF), Schottky, or Fowler–Nordheim (FN) conduction using BIC model selection plus a dynamic-permittivity admissibility test. The regime map conditions the prior windows and the loss weights, and—equally important—documents where the Stanford current law is structurally unable to follow the data.
2) *Measured-cycle anchoring.* The nominal fit target is a medoid cycle selected from the quality-controlled cycle ensemble, not a synthetic pointwise average, preserving measured hysteresis ordering and compliance behavior.
3) *Operational identifiability reporting.* Each fit is followed by Fisher analysis at the optimum and 200 cycle-level bootstrap refits, separating locally constrained coordinates from parameters that are unstable under measured cycle variability.
4) *A cross-condition case study.* Applied to twelve deposition conditions, the flow reproduces all measured butterfly curves to 0.47–0.77 (SET) and 0.65–1.02 (RESET) decades RMS while showing that the active parameter vector is severely ill-conditioned; it also identifies Schottky-limited high-resistance-state (HRS)

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TABLE I
DEPOSITION DESIGN OF EXPERIMENTS. CENTER POINT (27.5%, 162.5 W) IS REPLICATED FOUR TIMES (S9–S12).

Sample	O_2 (%)	Power (W)
S1	20	75
S2	20	250
S3	35	75
S4	35	250
S5	20	162.5
S6	35	162.5
S7	27.5	75
S8	27.5	250
S9–S12	27.5	162.5

conduction—absent from the Stanford model—as the dominant source of residual structure.

Section II describes the devices and measurement design. Section III summarizes the extraction pipeline. Section IV reports fits, identifiability, ablations, and variability across the twelve conditions. Section V discusses implications for compact-model users, and Section VI concludes.

II. Devices and Measurement Design

The devices are sputter-deposited TaO_x cross-point RRAM cells ($4\,\mu\mathrm{m}$ feature size) fabricated and characterized as part of an ongoing deposition-condition study [6], [7]. Twelve sample conditions (S1–S12) span a face-centered factorial design over oxygen percentage in the sputter ambient and RF sputter power, with four replicated center points (Table I). This design supports response-surface analysis of extracted parameters versus deposition conditions once per-condition parameter sets are available.

Bipolar DC switching was measured with a Keysight B1500 semiconductor parameter analyzer; raw data are CSV exports containing one or more SET, RESET, or mixed sweep blocks per file. The fitting current compliance is $I_{\mathrm{cc}} = 5\times10^{-4}$ A; because the Stanford Verilog-A model contains no instrument compliance circuit, simulated currents are clamped to this magnitude after each transient. An automated quality screen (minimum point count, voltage span, peak-current window, dynamic range, and SET+RESET completeness) rejects shorted, open, incomplete, or noise-floor sweeps before any fitting. For the representative S1 device, for example, all 40 measured cycles pass the screen with a mean quality score of 8.9/10.

III. Extraction Methodology

The pipeline comprises eight stages; this section summarizes each stage and its design rationale. A complete stage-by-stage specification, including the prior table and all configured constants, is provided in the released implementation.

A. Representative Medoid Cycle

For each condition, the fitted device is selected by ranked criteria (number of good cycles, mean quality score, good-cycle fraction), which prevents the fitting target from being chosen by visual preference. Programming segments are isolated per cycle and branch by splitting the voltage trace at direction reversals. All cycles are interpolated in $\log_{10}|I|$ onto a common voltage grid, and the cycle that minimizes the median absolute deviation from the branch-wise median log-current,

$$s_{c,b} = \mathrm{median}_v \left|\log_{10}|I_{c,b}(v)| - \mathrm{median}_{c'} \log_{10}|I_{c',b}(v)|\right|, \tag{1}$$

averaged over both branches, is retained as the *medoid* fitting target. Anchoring the point estimate to one measured sweep—rather than a synthetic pointwise median—preserves the measured hysteresis ordering, switching kinks, and compliance-limited portions that a compact model must actually reproduce. The full cycle ensemble is retained separately for the variability stage (Section III-H).

B. Conduction-Regime Classification

Each branch of the medoid cycle is split at its voltage extremum into pre/post-switching state segments (HRS and LRS sides of SET and RESET). Within sliding windows of each segment, four standard transport linearizations are regressed:

$$\text{ohmic:} \quad \ln I = m\ln V + b, \quad 0.8 \le m \le 1.2, \tag{2}$$
$$\text{PF:} \quad \ln(I/V) = s_{\mathrm{PF}}\sqrt{V} + b, \quad s_{\mathrm{PF}} > 0, \tag{3}$$
$$\text{Schottky:} \quad \ln I = s_{\mathrm{S}}\sqrt{V} + b, \quad s_{\mathrm{S}} > 0, \tag{4}$$
$$\text{FN:} \quad \ln(I/V^2) = s_{\mathrm{FN}}/V + b, \quad s_{\mathrm{FN}} < 0. \tag{5}$$

PF and Schottky linearizations are nearly collinear over short windows [8], so a physical admissibility test precedes model selection: the dynamic relative permittivity implied by the fitted $\sqrt{V}$ slope s,

$$\varepsilon_r^{\mathrm{dyn}} = \frac{q^3}{\pi\varepsilon_0 t_{\mathrm{ox}}(skT)^2} \times \begin{cases} 1, & \text{PF}, \\ 1/4, & \text{Schottky}, \end{cases} \tag{6}$$

must fall in the deliberately broad bracket $[1, 60]$ under the device oxide thickness $t_{\mathrm{ox}} = 7\,\mathrm{nm}$, $T = 300\,\mathrm{K}$. Admitted candidates are compared by the Bayesian information criterion; adjacent same-label windows are merged and refit. Windows labeled ohmic or PF are marked `stanford_valid`, since these are the behaviors the released Stanford current law can emulate; Schottky, FN, and unclassified windows are not. When the cycle ensemble is supplied, every cycle is classified at the state-segment level, yielding a cycle-to-cycle mechanism-stability table.

C. Compact Model and HSPICE Forward Map

The fitted model is the Stanford–PKU RRAM Verilog-A model `rram_v_1_0_0` [3], [4], simulated in HSPICE through generated transient netlists. The terminal current and deterministic gap dynamics are

$$I_{\mathrm{tb}} = I_0 \exp\left(-\frac{g}{g_0}\right)\sinh\left(\frac{V_{\mathrm{tb}}}{V_0}\right), \tag{7}$$

$$\frac{dg}{dt} = -\nu_0 \exp\left(-\frac{qE_a}{kT_{\mathrm{cur}}}\right)\sinh\left(\frac{\gamma a_0}{t_{\mathrm{ox}}}\frac{qV_{\mathrm{tb}}}{kT_{\mathrm{cur}}}\right), \tag{8}$$

with local temperature $T_{\mathrm{cur}} = T_{\mathrm{ini}} + |V_{\mathrm{tb}} I_{\mathrm{tb}} R_{\mathrm{th}}|$ and a gap-dependent field-enhancement factor γ. A series resistor R_s

models probe/parasitic resistance. Each objective evaluation runs two transients: a SET sweep initialized in HRS and a RESET sweep initialized in LRS, with polarity-split parameters (V_0, E_a, $F_{\min}$, ν_0, $g_{\max}$, g_{ini}) so the model can express measured SET/RESET asymmetry while scale and parasitic parameters remain shared. Measured voltage samples are replayed as a quasi-static piecewise-linear source; the fitted ν_0 and E_a therefore absorb the effective sweep-time scale and should be compared within this protocol rather than read as protocol-independent kinetic constants.

D. Regime-Conditioned Priors

Each of the 21 model parameters receives a prior mean, standard deviation, and hard physical bounds. When a regime map is available, the prior windows are selected from classified physics instead of fixed voltage ranges: the LRS current scale (I_0, V_0) is regressed as $I = A\sinh(V/V_0)$ over the classified LRS ohmic window, and—when a PF window exists—HRS PF slopes and intercepts map to γ_0 and E_a priors. When no PF window exists for a branch the code does not force a PF fit onto non-PF data; it retains literature-centered defaults ($\gamma_0 = 16.5$, $E_a = 0.6\,\text{eV}$) with inflated uncertainty and records the missing PF prior in a `pf_available` flag. As shown in Section IV-A, this honesty mechanism turns out to be essential for the present devices. The series-resistance prior is sized from the measured LRS resistance.

E. Mechanism-Aware Objective

The optimizer minimizes a variance-scaled log-current loss with switching threshold and prior penalties,

$$L(\boldsymbol{\theta}) = L_{\text{SET}} + L_{\text{RESET}} + 0.5L_V + 0.1L_{\text{prior}}, \qquad (9)$$

where each branch loss is a weighted mean of squared residuals $r_i = \left(\log_{10}|I_i^{\text{sim}}| - \log_{10}|I_i^{\text{meas}}|\right)/\sigma_i^{\log I}$, L_V penalizes the mismatch of switching thresholds located at the extrema of $d\log_{10}|I|/dV$, and the prior term is a quadratic penalty in prior-standardized coordinates. Points covered only by Schottky/FN windows receive weight $w_{\text{nb}} = 0.3$; ohmic/PF and unclassified points receive full weight. The down-weighting limits the ability of structurally unmodeled interface conduction to bias parameters that the Stanford law can represent, without hiding the mismatch.

F. Global Search and Local Refinement

Global search uses Bayesian optimization (expected-improvement-plus, 48 quasi-random seeds + 160 adaptive evaluations, fixed seed) over prior-scaled boxes, with log transforms for scale-like parameters [10]; 16 of the 21 parameters are active by default. The best point is polished by Nelder–Mead in log-parameter coordinates (100 evaluations) with out-of-bounds penalties. A nominal point estimate costs $2(48 + 160 + 100) = 616$ HSPICE branch transients.

G. Operational Identifiability

Identifiability is evaluated *at the fitted optimum*. A finite-difference sensitivity matrix in log-parameter coordinates yields the Fisher matrix $F = \widetilde{S}^{\mathsf{T}}\widetilde{S}$; because such parameterizations are sloppy, effective uncertainties are computed from the pseudo-inverse, $\sigma_j^{\text{eff}} = \sqrt{[F^{+}]_{jj}}$, and parameters are classed by relative uncertainty $\rho_j = \sigma_j^{\text{eff}}/|\hat{\theta}_j|$ (identifiable: $\rho < 0.5$; weak: $0.5 \le \rho < 2$; non-identifiable: $\rho \ge 2$) [5], [9]. The Fisher result is interpreted as a local diagnostic rather than a uniqueness proof: very large condition numbers, numerical null directions, and disagreement with cycle-bootstrap stability are all treated as evidence against assigning microscopic meaning to an individual coordinate.

H. Cycle-to-Cycle Variability

Cycle-to-cycle uncertainty uses 200 cluster-bootstrap draws over whole measured cycles of the selected device: each draw resamples cycles with replacement, collapses them to one median curve by branch and point index (preserving hysteresis sequence), and refits with a reduced budget ($8 + 24$ evaluations). Percentiles across draws give empirical 95% confidence intervals that reflect cycle-to-cycle variability under the fitting protocol; they make no claim about device-to-device or wafer-level populations.

I. Ablation Design

To isolate the contribution of each ingredient, every condition is also fitted under controlled variants: physics priors with Bayesian optimization (`physics_bo`), uninformative priors over the hard physical bounds (`plain_bo`), prior means without any optimization (`physics_nobo`), and the full regime-conditioned method (`physics_regime_bo`). The ablation reports *unweighted* validation RMSE so that methods using mechanism-aware training weights are not favored by their own loss.

IV. Results

A. Conduction Regimes: Ohmic LRS, Schottky-Limited HRS

Figure 1 shows the classified regime map for the representative S1 device, and Fig. 2 summarizes the cycle-level dominant mechanism for all twelve conditions. The post-SET LRS is ohmic-dominant for every condition, with stability from 66.3% (S2) to 100%. More consequentially, the post-RESET HRS is Schottky-dominant for all conditions in 92.5–100% of cycles. Its representative-cycle Schottky windows imply $\varepsilon_r^{\text{dyn}} = 4.72$–$33.96$ under $t_{\text{ox}} = 7\,\text{nm}$, inside the deliberately broad admissibility interval.

The pre-SET HRS is Schottky-dominant in ten conditions, with stability from 60.4% (S2) to 100% (S8); S5 and S7 are instead ohmic-dominant at 80% and 75%, respectively. The pre-RESET LRS is the least stable segment and is often unclassified, consistent with a transition between the conductive filament branch and rupture. PF windows occur only in the representative pre-SET maps of S6 and S11, and PF is not the dominant cycle-level mechanism for any state or condition.

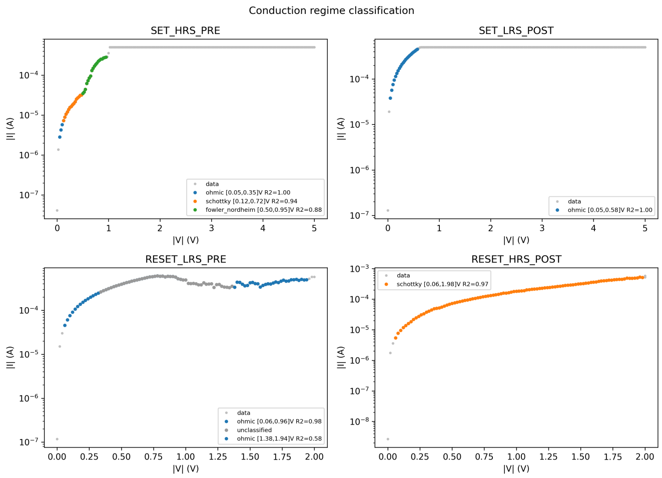

Fig. 1. Conduction-regime classification of the four resistance-state segments of the S1 representative cycle. LRS segments are ohmic ($R^2 \geq 0.98$); HRS segments are dominated by Schottky emission, with a Fowler–Nordheim window near the SET transition. No Poole–Frenkel window is admitted by the $\varepsilon_r^{\text{dyn}}$ test.

This result has two consequences for compact-model extraction. First, the PF-to-(γ_0, E_a) prior pathway fires only on the SET side of S6 and S11; all RESET-side and remaining SET-side activation-energy priors retain literature-centered defaults with inflated uncertainty. Second, a substantial fraction of the HRS sweep is covered only by `stanford_valid`=0 windows and enters the loss at weight 0.3. The Stanford current law (7) contains no interface-barrier term, so residual structure in these regions is expected and is quantified below.

B. Point-Estimate Fit Quality

Table II reports unweighted validation RMSE (in decades of log-current) and Durbin–Watson (DW) statistics for all twelve conditions; Fig. 3 shows the S1 overlay and residuals. SET-branch RMSE spans 0.47–0.77 decades (mean 0.530) and RESET-branch RMSE spans 0.65–1.02 decades (mean 0.761). The fitted models capture the hysteresis ordering, both switching transitions, the compliance plateau, and the LRS return branches; the dominant residual is a localized error of up to one decade at the SET transition and along the Schottky-limited HRS approach.

The DW statistics (0.46–1.04, all well below 2) quantify what the overlay plots suggest: residuals are strongly autocorrelated along the sweep. We report this deliberately. Structured residuals indicate systematic compact-model mismatch—consistent with the missing interface-conduction term identified in Section IV-A—rather than random measurement noise, and averaging them away with looser metrics would overstate model adequacy. S2, with the largest RESET RMSE (1.02 decades) and the lowest SET DW, also has the least stable Schottky assignment among the Schottky-dominant pre-SET conditions (60.4%), supporting this reading.

C. Identifiability: What DC Data Actually Constrain

Figure 4 shows the Fisher relative-uncertainty heatmap across all twelve conditions for the archived physics-prior

TABLE II
VALIDATION FIT QUALITY PER CONDITION: UNWEIGHTED RMSE OF $\log_{10}|I|$ (DECADES) AND DURBIN–WATSON STATISTICS.

Cond.	RMSE_{SET}	$\text{RMSE}_{\text{RESET}}$	DW_{SET}	DW_{RESET}
S1	0.500	0.689	0.86	0.85
S2	0.772	1.016	0.46	0.56
S3	0.566	0.659	0.72	0.98
S4	0.514	0.730	0.82	0.90
S5	0.508	0.805	0.85	0.74
S6	0.468	0.650	0.94	0.93
S7	0.477	0.726	1.04	0.91
S8	0.495	1.001	1.03	0.54
S9	0.510	0.700	0.92	1.04
S10	0.508	0.718	1.00	0.98
S11	0.494	0.753	0.92	0.88
S12	0.548	0.681	0.96	1.01
Mean	0.530	0.761	–	–

BO point estimates. This audit uses the same 16-active-parameter model and data as the full method; the regime-conditioned arm is evaluated separately in the ablation and bootstrap studies. Taken at face value, 0–8 of the 16 active parameters per condition classify as Fisher-identifiable, with a mean of 2.4. The Fisher condition numbers range from 6.2×10^{18} to 1.6×10^{66}, confirming an extremely sloppy parameterization whose pseudo-inverse uncertainties must be interpreted cautiously.

Series resistance is the most recurrent nominally identifiable coordinate (8 of 12 conditions), followed by $F_{\text{min,RESET}}$ (5 of 12) and $V_{0,\text{SET}}$, $V_{0,\text{RESET}}$, and $F_{\text{min,SET}}$ (4 of 12 each). Neither I_0 nor g_0, and none of the four fitted gap-limit/initialization parameters, is Fisher-identifiable on any condition. Isolated identifiable classifications for activation energy or velocity prefactor occur only on S9 or S12 and are not stable across the replicated center-point conditions S9–S12. Extremely small pseudo-inverse uncertainties on individual F_{min} entries are therefore treated as local numerical results, not as proof of global uniqueness.

The defensible interpretation is that DC butterfly data constrain effective branch scales—series resistance, voltage curvature, and switching-field combinations—more reliably than the decomposition into I_0, g_0, activation energies, velocity prefactors, and gap geometry. The bootstrap results in Section IV-E independently reinforce this conclusion by showing substantial parameter movement under resampling of the measured cycles.

D. Ablation: What Each Ingredient Buys

Table III compares the extraction variants on unweighted validation RMSE averaged over all twelve conditions. Bayesian optimization is decisive on RESET: physics-informed prior means without optimization (`physics_nobo`) yield 1.528 ± 0.434 decades, versus 0.756 ± 0.124 with BO. SET is less sensitive because its LRS and compliance-limited portions are already close to the prior prediction.

The three optimized variants have nearly identical aggregate unweighted error. The full regime-aware method gives

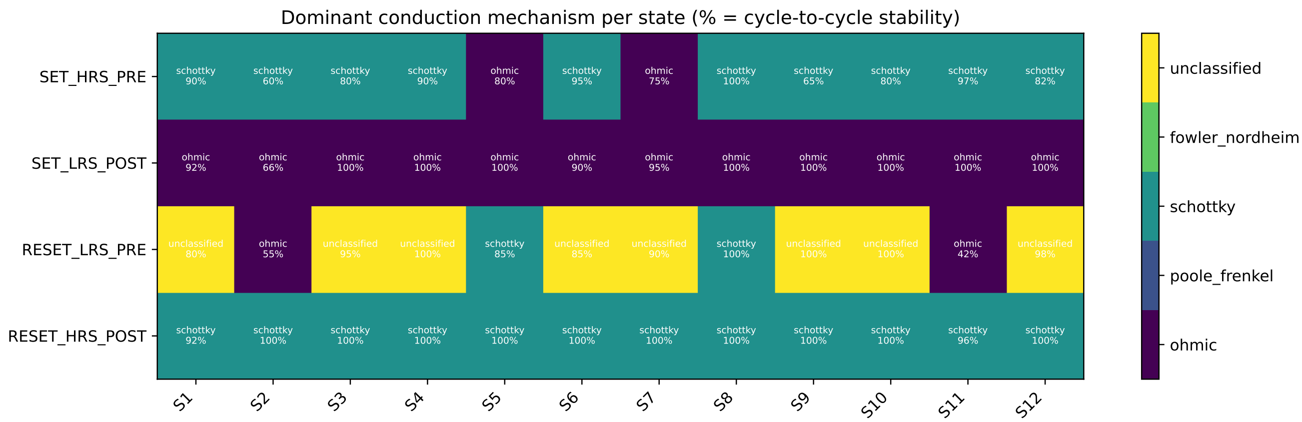

Fig. 2. Dominant conduction mechanism and cycle-to-cycle assignment stability for each resistance-state segment and deposition condition. The post-SET LRS is consistently ohmic, whereas the post-RESET HRS is Schottky-dominant for all twelve conditions.

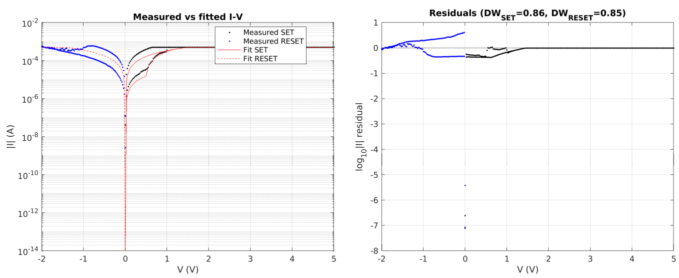

Fig. 3. Measured versus fitted I–V for condition S1 (left) and log-current residuals along the sweep (right). The fit reproduces the butterfly hysteresis, both transitions, and the compliance plateau; the residual concentrates at the SET transition and the Schottky-limited HRS approach.

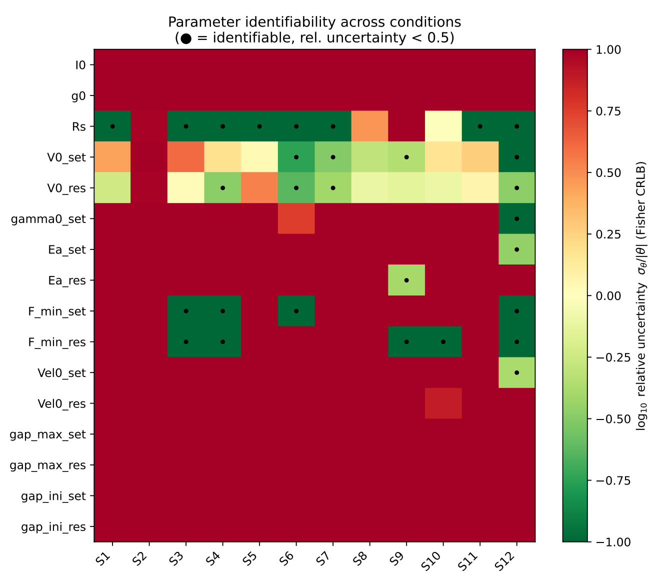

Fig. 4. Fisher relative uncertainty $\sigma_\theta/|\theta|$ (log color scale) for the 16 active Stanford parameters across the twelve conditions; dots mark the nominal identifiable class ($\rho < 0.5$). The very large condition numbers require these local classifications to be interpreted together with cycle-bootstrap stability.

TABLE III
ABLATION OVER ALL 12 CONDITIONS: UNWEIGHTED VALIDATION RMSE (DECADES), MEAN ± STANDARD DEVIATION ACROSS CONDITIONS.

Variant	$\mathrm{RMSE}_{\mathrm{SET}}$	$\mathrm{RMSE}_{\mathrm{RESET}}$
Regime-conditioned + weighting	0.530 ± 0.081	0.761 ± 0.123
Physics priors + BO	0.546 ± 0.077	0.756 ± 0.124
Uninformative priors + BO	0.537 ± 0.076	0.755 ± 0.126
Physics priors, no BO	0.571 ± 0.115	1.528 ± 0.434

0.530/0.761 decades, legacy physics priors give 0.546/0.756, and uninformative priors give 0.537/0.755. Thus, regime conditioning and informative priors should not be sold as an RMSE improvement. Their role is to prevent unsupported Schottky/FN windows from steering a bulk-current model and to document how poorly identified coordinates are selected. In a sloppy model, comparable errors from different prior structures are themselves evidence that an overlay is insufficient to establish physical parameter extraction.

E. Cycle-to-Cycle Variability

The cluster bootstrap of Section III-H propagates cycle-to-cycle variability of the selected device into empirical 95% intervals for every active parameter. All 2400 refits (200 draws for each of 12 conditions) completed, and all 192 condition–parameter intervals are finite and nondegenerate. Figure 5 reports the coefficient of variation (CV) of each bootstrap distribution. The median CV over all entries is 0.448, with a range of 0.202–2.698.

The largest cross-cycle instability occurs in R_s (median CV 1.444) and the SET/RESET velocity prefactors (1.194 and 1.282). Current scale I_0 and activation energies also vary substantially (median CV 0.566 and 0.456–0.591). The branch voltage scales are more repeatable, with median CV near 0.28, consistent with the switching-curve shape directly constraining them. Some gap parameters show lower CV despite failing the Fisher test; this is not contradictory because a bounded, prior-centered optimizer can return a stable coordinate even when the likelihood supplies little local information. Boot-

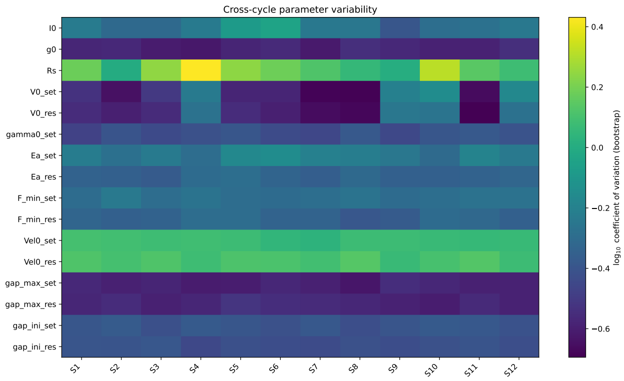

Fig. 5. Cycle-to-cycle parameter variability from 200 cluster-bootstrap refits per condition. Color denotes $\log_{10}(\mathrm{CV})$. Series resistance and velocity prefactors are the least stable, while the branch voltage scales are comparatively repeatable.

strap repeatability and identifiability therefore answer different questions and must be reported together.

V. Discussion

A. Implications for Compact-Model Users

For circuit designers consuming calibrated Stanford parameter sets, the practical messages are direct. (i) Treat per-condition parameter files as *behavioral* calibrations valid for the fitted protocol; only the identifiable combinations transfer with evidential support. (ii) Do not ascribe physical meaning to fitted E_a, ν_0, or gap geometry obtained from DC sweeps alone; constraining kinetics requires pulsed or sweep-rate-resolved data, which break the quasi-static degeneracy between kinetic prefactors and the time axis. (iii) When comparing parameters across deposition conditions (the response-surface use case of Table I), compare identifiable combinations—LRS resistance scale, switching thresholds—rather than raw parameter columns, since the latter mix data- and prior-driven coordinates.

B. Model Adequacy for TaO_x

The regime analysis localizes the Stanford model's structural limitation for these devices: HRS conduction is interface-limited (Schottky) in nearly all cycles, while the model's single $\sinh$ current law is bulk-like. This mismatch, not optimizer failure, sets the residual floor of Table II—the three optimized ablation arms have nearly identical aggregate unweighted RMSE despite different priors and loss weighting. A principled extension would add a barrier-limited HRS current branch (or a gap-dependent barrier term); the regime map produced here provides exactly the windows and slopes such an extension should target. Until then, down-weighting unmodelable windows during fitting, as done here, prevents interface physics from contaminating bulk parameters while keeping the mismatch visible in unweighted validation.

C. Limitations

The transient replay is quasi-static, so kinetic parameters absorb the effective time scale. Fisher analysis is local, and the present result archive does not contain completed profile-likelihood scans; identifiability claims are therefore limited to the combined Fisher and bootstrap evidence. The Fisher audit and full regime-conditioned bootstrap were also archived as separate runs, so a future release should repeat both diagnostics at one common optimum. Bootstrap intervals are cycle-level for one selected device per condition and do not constitute device-to-device or wafer-level statistics. Finally, the reset-side γ_0 is structurally overridden under negative bias in the released Verilog-A, so it is retained for model compatibility but not physically interpreted.

VI. Conclusion

We presented a physics-guided, identifiability-aware pipeline that fits the Stanford RRAM compact model to measured TaO_x DC switching data and reports, with every parameter set, the evidence behind it. Applied uniformly to twelve sputter-deposition conditions, the flow achieves mean RMSE of 0.530 decades for SET and 0.761 decades for RESET while establishing three findings that overlay plots alone cannot. First, only 0–8 of 16 active parameters per condition pass the local Fisher threshold, and the matrices remain extremely ill-conditioned. Second, 200 cycle-level bootstrap refits per condition reveal substantial parameter variability, particularly in series resistance and velocity prefactors. Third, the post-RESET HRS is Schottky-dominant in 92.5–100% of cycles for every condition, a mechanism outside the Stanford bulk current law that bounds achievable fit fidelity. Fit quality, regime validity, identifiability, and cycle-level uncertainty should therefore be reported together as the deliverable of RRAM compact-model calibration.

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